Quiz — 2.1.4 Thinking Logically

Name: ______________________ Date: __________ Mark: ______ / 20

OCR H446 · Component 02 · 2.1.4

Section A — Multiple choice (6 marks)

1. In a program, a *decision point* is a place where: A. Data is stored in fast memory for reuse B. The flow of control branches depending on a condition C. A problem is broken into sub-procedures D. Two tasks run at the same time Your answer: ☐

2. A cinema offers a discount if the customer is under 16. The correct Boolean condition for the decision point is: A. `IF age = 16` B. `IF age < 16` C. `IF age >= 16` D. `IF age > 16` Your answer: ☐

3. A booking system confirms a seat only when `seatsRequested <= seatsAvailable`. What happens on the **FALSE** branch of this decision? A. The seats are booked successfully B. The booking is rejected / the user is told there are too few seats C. The seats are cached D. Nothing — there is no false branch Your answer: ☐

4. A thermostat turns the heating ON when the temperature is below 18°C, otherwise OFF. How many outcomes (branches) does this single decision point have? A. None B. One C. Two D. Four Your answer: ☐

5. A loop "repeat while there are more records to process" relies on a condition tested each iteration. This loop test is an example of: A. A reusable component B. A decision point controlling flow of control C. Abstraction D. A precondition that is only checked once Your answer: ☐

6. A login system locks the account after 3 failed attempts. The decision should fire when: A. `IF attempts = 1` B. `IF attempts >= 3` C. `IF attempts < 3` D. `IF attempts <= 0` Your answer: ☐

Section B — Short answer (9 marks)

7. A game awards a bonus life when a player's score reaches 1000 or more. Write the exact Boolean **condition** for this decision point and state the outcome of **both** branches. **[3]**

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8. A car park barrier opens only if a ticket is valid **and** the fee has been paid. Write a suitable Boolean condition using a logical operator, and state what happens when the condition is false. **[2]**

9. Explain what is meant by *flow of control* and how decision points affect it, using a simple example. **[2]**

10. A shop offers free delivery when the basket total is over £50. A student writes `IF total > 50`. State whether a £50 order qualifies under this condition, and rewrite the condition if free delivery should include exactly £50. **[2]**

Section C — Applied question (5 marks)

11. A streaming service decides which version of a film to play. If the user has a premium subscription it streams in 4K; otherwise it streams in HD, unless the user's connection is slow (below 5 Mbps), in which case it always streams in SD regardless of subscription. Using *thinking logically*, identify the decision points, write exact Boolean conditions for each, and state the outcome of each branch. **[5]**